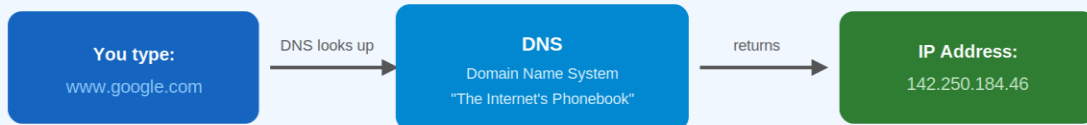


Networking Fundamentals

Domain Name System

What is DNS

- DNS (Domain Name System) is the Internet's phonebook
- Translates human-readable domain names into IP addresses
- Example: google.com → 142.250.184.46
- Enables browsers to find and load websites



Your browser uses the IP address to load the website — you never have to memorise numbers.

Anatomy of a Domain Name

Domain names are made of labels separated by dots. The hierarchy is read from right to left.

Example: www.shop.example.com

- **com:** Top-Level Domain (TLD), the highest visible level
- **example:** Second-Level Domain (SLD), usually the registered brand name
- **example.com:** Root/Apex domain, the base for all DNS records
- **shop:** Subdomain, a child section of the main domain
- **www:** Hostname or common subdomain used for the website
- **www.shop.example.com:** FQDN (Fully Qualified Domain Name)

www.shop.example.com — Anatomy of a Domain Name



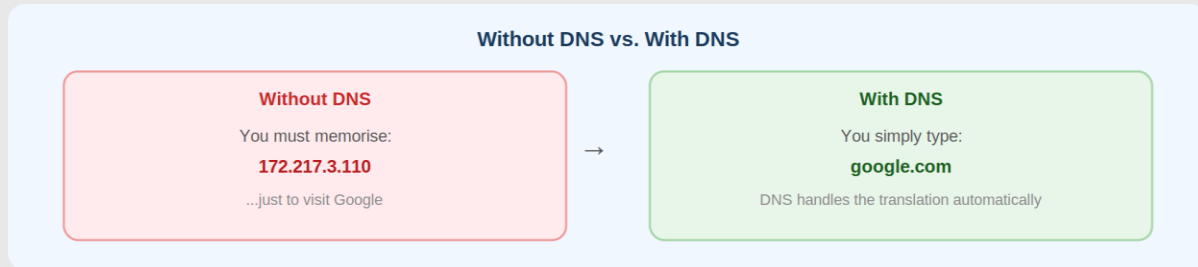
example.com = Root / Apex Domain

www.shop.example.com = Fully Qualified Domain Name (FQDN)

Domain hierarchy is read right to left: TLD → SLD → Subdomain → Hostname

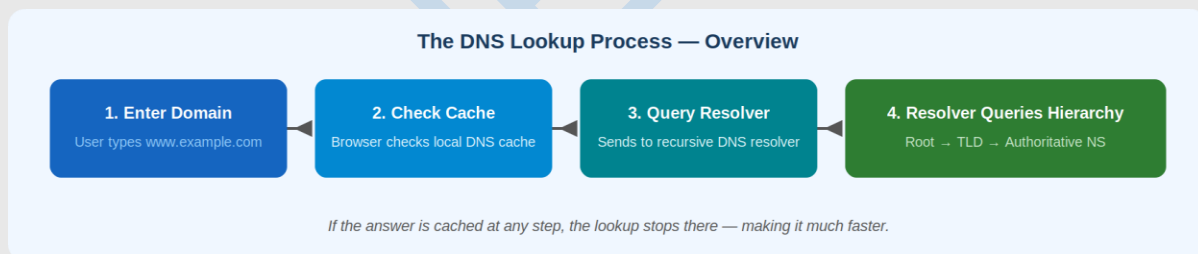
Why Do We Need DNS

- Humans prefer memorable names over numbers
- IP addresses are hard to remember (e.g. 172.217.3.110)
- Domain names are easy to remember (e.g. google.com)
- DNS makes the internet user-friendly and practical



The DNS Lookup Process

- User enters a domain in the browser (e.g. www.example.com)
- Browser checks local cache for cached DNS data
- If not found, sends a query to a recursive DNS resolver
- Resolver queries a hierarchy of DNS servers



DNS Hierarchy

1. Recursive Resolver

- Usually your ISP
- Acts as intermediary between your device and other DNS servers

2. Root Nameserver

- Points to TLD servers
- 13 sets globally, maintained by ICANN

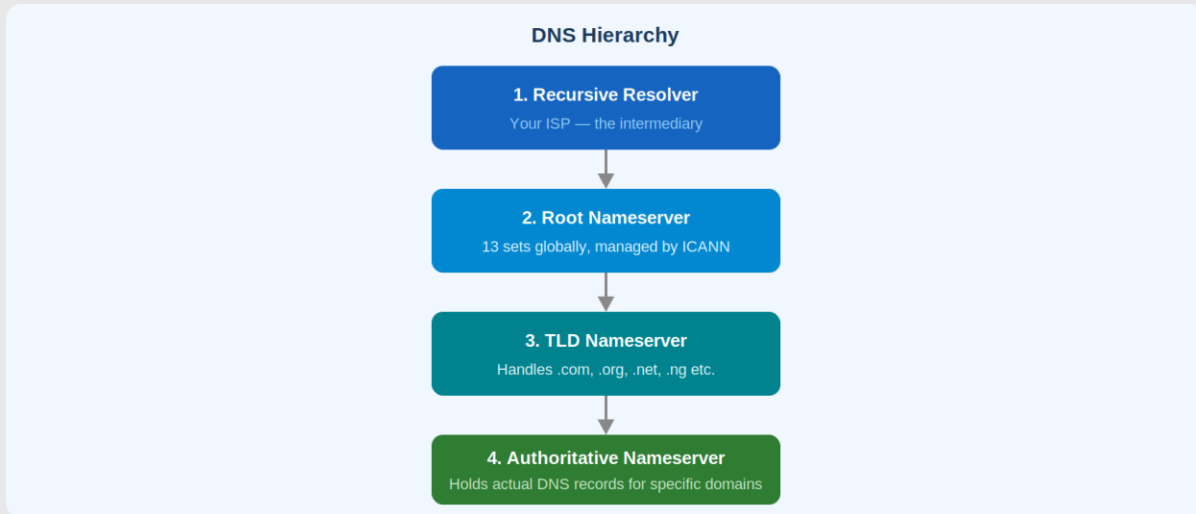
3. TLD Nameserver

- Top-Level Domain server

- Stores information for .com, .org, .net etc.

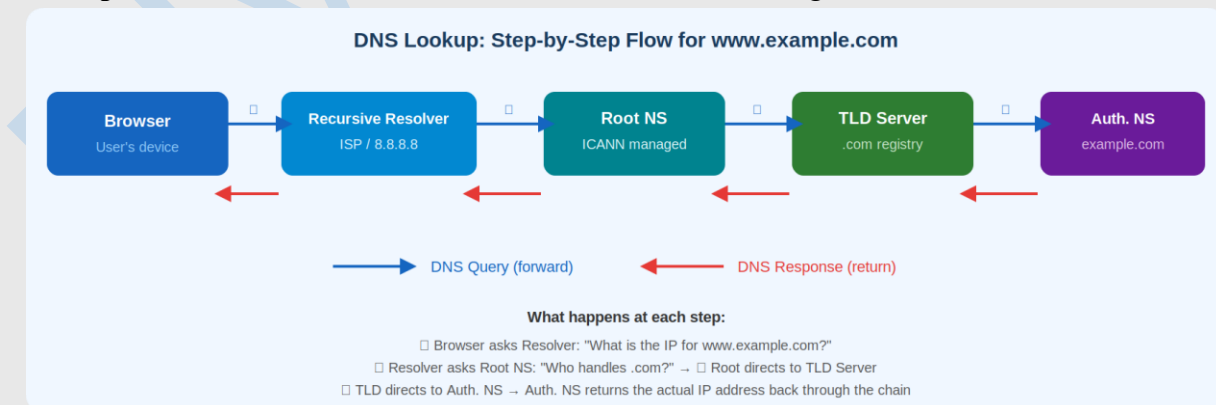
4. Authoritative Nameserver

- Holds the actual DNS records for specific domains
- The final stop, returns the real IP address



DNS Lookup: Step by Step

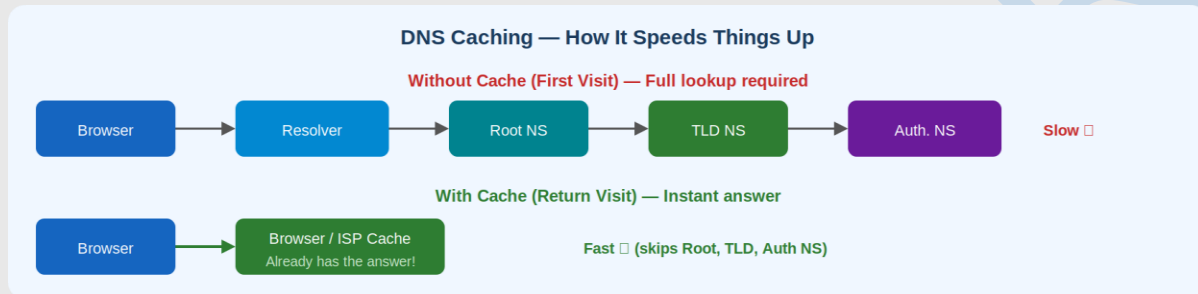
- **Step 1:** User types www.example.com in the browser
- **Step 2:** Browser queries the recursive resolver (ISP DNS)
- **Step 3:** Resolver queries the root nameserver for .com
- **Step 4:** Root server directs to the TLD server
- **Step 5:** TLD server directs to the authoritative nameserver for example.com
- **Step 6:** Authoritative NS returns the IP address back through the chain to the browser



DNS Caching: Speed Optimization

- DNS results are cached at multiple levels to avoid repeating full lookups
- Browser cache stores recently visited domains on your device
- ISP cache caches DNS queries for all its customers
- Cached results skip multiple lookup steps, making browsing much faster

Note: Caching is controlled by TTL (Time To Live), a value set by the domain owner that tells DNS servers how long to keep the answer.



DNS Record Types

A Record

- The most important DNS record
- Maps a domain name to an IPv4 address
- Example: example.com → 93.184.216.34

AAAA Record

- IPv6 equivalent of the A record
- Maps a domain name to an IPv6 address (128-bit)
- Example: example.com → 2606:2800:220:1:248:1893:25c8:1946

CNAME Record

- Alias for domain names
- Points one domain to another domain
- Example: blog.example.com → example.com

MX Record (Mail Exchange)

- Directs email to the correct mail server

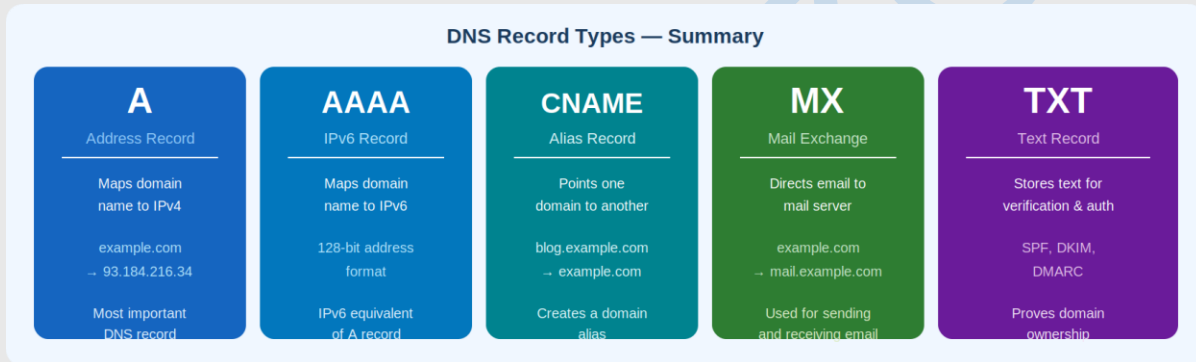
- Example: example.com → mail.example.com

NS Record (Nameserver)

- Delegates DNS authority to nameservers
- Example: example.com NS ns1.example.com

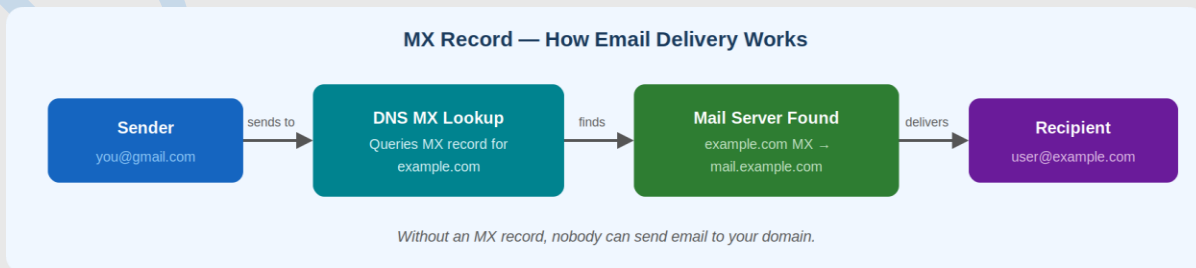
TXT Record

- Stores text for verification and authentication
- Uses: SPF, DKIM, DMARC for email security
- Used to prove domain ownership and authenticate services



Real-World example: Sending email to user@example.com

- Mail server queries the MX record for example.com
- Discovers that mail.example.com is the mail server
- Queries the A record of mail.example.com
- Gets IP address 192.168.1.10 and delivers the email

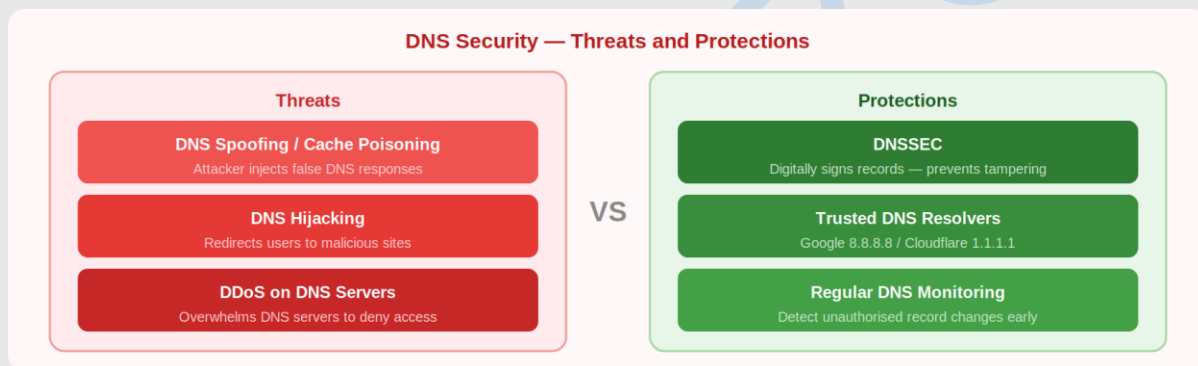


DNS Records Summary

- A: Maps domain to IPv4
- AAAA: Maps domain to IPv6
- CNAME: Creates a domain alias
- MX: Directs to mail server
- TXT: Text verification and authentication

DNS Security Considerations

- DNSSEC: Protects against DNS hijacking by digitally signing records
- Use trusted DNS resolvers such as Google (8.8.8.8) and Cloudflare (1.1.1.1)
- Be aware of DNS spoofing attacks: attackers can inject false DNS responses
- Regularly monitor DNS records to detect unauthorized changes early



Key Takeaways

- DNS translates domain names to IP addresses. It's like a phonebook for the internet
- The lookup process involves a Recursive Resolver, Root NS, TLD NS, and Authoritative NS
- Different record types serve different purposes: A, AAAA, CNAME, MX, TXT
- Caching makes DNS lookups significantly faster on repeat visits
- DNS security (DNSSEC, trusted resolvers, monitoring) is essential in the cloud